

$$1. (2) \quad |S_{n+p} - S_n| = \sum_{n+1}^{n+p} \frac{1}{\sqrt{n}} = \frac{1}{\sqrt{n+1}} + \frac{1}{\sqrt{n+2}} + \dots + \frac{1}{\sqrt{n+p}} \geq \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{n+p}$$

$$\text{令 } p=n+1, \text{ 有 } \sum_{n+1}^{2n+1} \frac{1}{\sqrt{n}} \geq \frac{1}{n+1} + \dots + \frac{1}{2n+1} \leq (n+1) \frac{1}{n+1} = 1$$

故级数发散

2. 不收敛

由于 $\sum_{n=1}^{\infty} a_n$ 收敛, 则 $\lim_{n \rightarrow \infty} a_n = 0$

而 $\sum_{n=1}^{\infty} b_n$ 发散, 则 $\lim_{n \rightarrow \infty} b_n \neq 0$

则 $\lim_{n \rightarrow \infty} (a_n \pm b_n) \neq 0$,

故级数一定发散

$$3. (1) \quad S_n = \sqrt{n+1} - 1 \quad \lim_{n \rightarrow \infty} S_n = \infty, \text{ 故发散.}$$

$$(4) \quad \lim_{n \rightarrow \infty} \cos^2 \frac{2}{n} = \cos^2 0 = 1, \text{ 发散} \quad \frac{1 + \cos \frac{2}{n}}{2}$$

$$(6) \quad \sum_{n=1}^{\infty} \left(\frac{\ln 5}{3}\right)^n, \text{ 而 } \frac{\ln 5}{3} < 1, \text{ 故收敛}$$

$$5. \because \sum_{n=1}^{\infty} u_n \text{ 收敛, 则 } \forall \varepsilon > 0, \exists N > 0, \left| \sum_{k=n+1}^{n+p} u_k \right| < \varepsilon$$

$$\text{即任取 } p \geq 1, \text{ 都有 } \lim_{n \rightarrow \infty} \sum_{k=n+1}^{n+p} u_k = 0$$

$$\text{令 } p=n, \text{ 即有 } \lim_{n \rightarrow \infty} (u_{n+1} + u_{n+2} + \dots + u_{2n}) = 0$$

$$\text{而 } u_{n+1} + u_{n+2} + \dots + u_{2n} \geq n \cdot u_{2n} \geq 0.$$

$$\text{由夹逼定理, } \lim_{n \rightarrow \infty} n \cdot u_{2n} = 0, \text{ 则 } \lim_{n \rightarrow \infty} 2n \cdot u_{2n} = 0$$

$$\text{而 } (2n+1)u_{2n+1} = 2n \cdot u_{2n+1} + u_{2n+1} \leq 2n \cdot u_{2n} + u_{2n+1},$$

$$0 \leq \lim_{n \rightarrow \infty} (2n+1)u_{2n+1} \leq \lim_{n \rightarrow \infty} (2n \cdot u_{2n} + u_{2n+1}) = 0$$

$$\text{故 } \lim_{n \rightarrow \infty} (2n+1)u_{2n+1} = 0$$

$$\text{故 } \lim_{n \rightarrow \infty} n \cdot u_n = 0$$

1. (3) 发散 $\frac{1}{\sqrt{n}} \geq \frac{1}{n}$, 而 $\sum_{n=1}^{\infty} \frac{1}{n}$ 发散, 故 $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ 发散

(5) 收敛 $\frac{n^n}{(n^2+3n+1)^{\frac{n+1}{2}}} \leq \frac{n^n}{n^{n+1}} = \frac{1}{n^2}$ 而 $\frac{1}{n^2}$ 收敛

故收敛
(6) 收敛 $(\ln n)^{\ln n} = e^{\ln n \cdot \ln \ln n} = n^{(\ln \ln n)}$ $\lim_{n \rightarrow \infty} \frac{n^{\ln(\ln n)}}{n^3} = \lim_{n \rightarrow \infty} n^{\ln(\ln n) - 3} = \infty$

当 n 充分大时, $(\ln n)^{\ln n} > n^3$

故 $\exists N > 0$, 当 $n > N$ 时, $\frac{n}{(\ln n)^{\ln n}} < \frac{n}{n^3} = \frac{1}{n^2}$, 而 $\sum_{n=2}^{\infty} \frac{1}{n^2}$ 收敛

故收敛

$$2. (3) a_n = \frac{3^n \cdot n!}{n^n} \quad \frac{a_n}{a_{n-1}} = 3n \cdot \frac{(n-1)^{n-1}}{n^n} = 3 \cdot \left(\frac{n-1}{n}\right)^{n-1} \quad \lim_{n \rightarrow \infty} \left(\frac{n-1}{n}\right)^{n-1} = \lim_{n \rightarrow \infty} e^{(n-1)\ln \frac{n-1}{n}} = \lim_{n \rightarrow \infty} e^{\frac{1-n}{n}} = e^{-1}$$

$$\therefore \lim_{n \rightarrow \infty} \frac{a_n}{a_{n-1}} = \frac{3}{e} > 1,$$

故发散

$$(5) a_n = \frac{n^2}{(3 - \frac{1}{n})^n}, \quad \sqrt[n]{a_n} = \frac{n^{\frac{2}{n}}}{3 - \frac{1}{n}}, \quad \lim_{n \rightarrow \infty} \sqrt[n]{a_n} = \frac{1}{3}, \text{ 故}$$

原级数收敛

$$(10) f(n) = \frac{1}{n \cdot (\ln n)^p} (n \geq 2), \quad f(n) \text{ 单调}$$

$$\int_2^{\infty} f(n) dn = \lim_{A \rightarrow \infty} \int_2^A \frac{1}{n (\ln n)^p} dn = \lim_{A \rightarrow \infty} \int_2^A \frac{1}{(\ln n)^p} d(\ln n) = \lim_{A \rightarrow \infty} \left[\frac{1}{1-p} (\ln n)^{1-p} \right] \Big|_2^A = \lim_{A \rightarrow \infty} \frac{1}{1-p} [(\ln A)^{1-p} - (\ln 2)^{1-p}]$$

① 若 $0 < p < 1$, $\int_2^{\infty} f(n) dn$ 不收敛, 级数发散

② 若 $p \geq 1$, $\int_2^{\infty} f(n) dn = \frac{(\ln 2)^{1-p}}{p-1}$, 级数收敛

3. 证明: 若正项级数 $\sum_{n=1}^{\infty} u_n$ 收敛, 则级数 $\sum_{n=1}^{\infty} u_n^2$ 也收敛. 反之不一定成立, 试举例说明

$\because \sum_{n=1}^{\infty} u_n$ 收敛, $\lim_{n \rightarrow \infty} u_n = 0$, 则 $\exists N > 0$, 当 $n > N$ 时, $0 < u_n < 1$

则必有 $n > N$ 时, $u_n^2 < u_n$

故 $\sum_{n=1}^{\infty} u_n^2$ 收敛

反例: $\sum_{n=1}^{\infty} \frac{1}{n^2}$ 收敛

而 $\sum_{n=1}^{\infty} \frac{1}{n}$ 发散

6. $\lim_{n \rightarrow \infty} \frac{u_n}{1/n} = l \in (0, +\infty)$, 而级数 $\sum_{n=1}^{\infty} \frac{1}{n}$ 发散, 则 $\sum_{n=1}^{\infty} u_n$ 发散

$\because \lim_{n \rightarrow \infty} n u_n = l$, 则 $\exists N > 0$, 当 $n > N$ 时, $\forall \varepsilon > 0$, $|n u_n - l| < \varepsilon$

$|n u_n - l| > |n u_n| - |l|$, $\therefore |n u_n| < \varepsilon + |l|$

取 $\varepsilon = \frac{|l|}{2}$, $|n u_n| < \frac{3|l|}{2}$.

$$n^2 u_n^2 < \frac{9l^2}{4}$$

$\therefore u_n^2 < \frac{9l^2}{4n^2}$, 级数 $\sum_{n=1}^{\infty} \frac{9l^2}{4n^2}$ 收敛

故级数 $\sum_{n=1}^{\infty} u_n^2$ 收敛